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Activity Title: Ambulance Routing and Traffic Signal Priority Using the Chinese Remainder

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ABSTRACT

Traffic congestion is one of the major problems faced by emergency vehicles, especially ambulances. During medical emergencies, even a small delay in reaching the hospital can have serious consequences for the patient.

One of the major causes of delay is the presence of traffic signals along the ambulance route. Since different traffic signals generally operate using different cycle times and starting offsets, an ambulance may have to stop repeatedly at red signals.

This project presents a mathematical approach for creating a **green corridor for ambulances** using the **Chinese Remainder Theorem (CRT)**. The Chinese Remainder Theorem is an important concept from number theory that is used to solve a system of simultaneous modular congruences.

In this project, each traffic signal is represented using a modular congruence based on its cycle time and required green-light timing.

The proposed method calculates the earliest time at which the traffic signals along a selected route can be synchronized so that the ambulance encounters green signals sequentially.

The cycle time of each traffic signal is treated as a modulus, while the required green time, adjusted according to the ambulance's travel time, is treated as a remainder. CRT is then applied to determine the common starting time for the ambulance.

The proposed mathematical model can help reduce unnecessary stops, decrease ambulance travel time, improve road safety, and provide a simple method for emergency-route planning.

The concept can also be extended to smart-city traffic management, IoT-based traffic signals, GPS-based ambulance tracking, fire brigade routes, police emergency routes, and disaster-relief operations.

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INTRODUCTION

Emergency medical transportation is a critical part of healthcare services. When a patient requires immediate medical attention, an ambulance must reach the hospital as quickly and safely as possible. However, in cities and highly populated areas, traffic congestion and traffic signals can cause significant delays.

An ambulance travelling through a city may encounter several traffic signals. Each traffic signal normally operates according to its own cycle time and starting offset. Therefore, all signals may not turn green at the same time. As a result, an ambulance may be forced to stop at multiple junctions during its journey.

The project "Ambulance Routing and Traffic Signal Priority Using the Chinese Remainder Theorem (CRT)" proposes a mathematical solution to this problem. The main idea is to model the operation of traffic signals using modular arithmetic and apply the Chinese Remainder Theorem to determine a suitable starting time for the ambulance.

The PPT identifies that independent traffic signals can force an ambulance to stop repeatedly and proposes CRT-based synchronization to create a green corridor.

- **Problem Statement**

Traffic signals along an ambulance route operate independently, with different cycle times and starting offsets. Because of this, an ambulance may encounter several red signals during its journey, resulting in repeated stops and increased travel time.

The problem is to develop a mathematical method that can determine the earliest suitable time at which an ambulance can start its journey so that the required traffic signals become green when the ambulance reaches them.

The project uses the Chinese Remainder Theorem to represent the traffic signals as modular congruences and calculate the common time required for synchronization.

- **Need of the System**

The system is required because ambulance delays can negatively affect emergency medical services. A conventional traffic system treats most vehicles equally and does not necessarily provide a coordinated green path for an emergency vehicle.

The proposed system provides a mathematical method for coordinating traffic signals on a selected ambulance route.

The major needs are:

1. To reduce ambulance travel time.
2. To reduce unnecessary stops at traffic signals.
3. To provide a mathematically calculated green corridor.
4. To improve emergency transportation.
5. To reduce the risk associated with stopping at busy junctions.
6. To provide a simple mathematical model for traffic-signal synchronization.
7. To support future smart-city traffic management systems.

- **Objectives**

The main objectives of the project are:

- To reduce the time required by an ambulance to reach the hospital.
- To represent every traffic signal using a modular congruence.
- To apply the Chinese Remainder Theorem for finding a common green time.
- To create a green corridor for emergency vehicles.
- To reduce unnecessary stopping at traffic junctions.
- To improve safety for patients, ambulance drivers, and other road users.
- To demonstrate a practical application of Discrete Mathematics and Number Theory.

LITERATURE SURVEY

- **Introduction**

A literature survey is an important part of a technical project because it helps in understanding existing concepts, methods, and mathematical techniques related to the proposed system.
For this project, the literature survey focuses mainly on **traffic signal synchronization, emergency vehicle priority, modular arithmetic, and the Chinese Remainder Theorem**.
The basic idea behind the proposed system is that traffic signals operate periodically. Since each signal has a particular cycle, its green-light condition can be represented using modular arithmetic.

- **Traffic Signal Synchronization**

Traffic signal synchronization is a technique used to coordinate multiple traffic signals so that vehicles can move through a sequence of intersections with fewer stops.
Traditional traffic signals generally operate according to predefined timing plans. However, when different signals have different cycle lengths and starting offsets, synchronization becomes more difficult.
For emergency vehicles such as ambulances, synchronization becomes especially important because stopping at multiple traffic signals can increase travel time.
The proposed project models each signal mathematically rather than treating the traffic system only as a physical infrastructure problem.

- **Emergency Vehicle Priority**

Emergency vehicle priority refers to giving special consideration to vehicles such as ambulances, fire trucks, and police vehicles.
An ambulance may need to travel through several intersections before reaching a hospital. If the ambulance encounters red signals repeatedly, its journey may become longer.
A green corridor can reduce these interruptions by coordinating traffic signals along the route.
The proposed project focuses specifically on creating such a green corridor using a mathematical approach.

- **Modular Arithmetic**

Modular arithmetic is a mathematical system based on remainders.
For example:

$$x \equiv 2 \pmod 3$$

means that when $x$is divided by 3, the remainder is 2.
Traffic signals have repeating cycles, which makes modular arithmetic useful for representing their behavior.
For example, if a traffic signal becomes green at a particular time and repeats after every $m$units, its green condition can be represented as:

$$x \equiv a \pmod m$$

where:

- x= required time,
- a= required green-time remainder,
- m= signal cycle time.

- **Chinese Remainder Theorem**

The Chinese Remainder Theorem is a fundamental theorem of number theory used to solve systems of simultaneous congruences.
For pairwise coprime moduli, a system such as:

$$x \equiv a_1 \pmod{m_1}$$
$$x \equiv a_2 \pmod{m_2}$$
$$x \equiv a_3 \pmod{m_3}$$

has a unique solution modulo:

$$N = m_1 m_2 m_3$$

SYSTEM DESIGN

- **System Overview**

The proposed system consists of a route containing multiple traffic signals through which an ambulance needs to travel.
Each signal has:

- A cycle time.
- A starting offset.
- A green-light condition.
- A specific position on the ambulance route.
- A travel time from the ambulance starting point.

The system converts this information into modular congruences and applies CRT to calculate the appropriate ambulance starting time.

- **Mathematical Model**

Each traffic signal is represented using the following congruence:

$$x \equiv a_i \pmod{m_i}$$

- **System Components**

1. Ambulance
The ambulance is the emergency vehicle for which the green corridor is created.
2. Traffic Signals
Each traffic signal has an independent cycle time.
3. Route Information
The system requires the route followed by the ambulance.
4. Travel Time
Travel time between consecutive signals is required so that the green-light condition can be adjusted.
5. Mathematical Processing
The collected information is converted into modular equations.
6. CRT Calculation
The Chinese Remainder Theorem is applied to obtain the common starting time.
7. Green Corridor
The calculated time is used as the ambulance's starting time so that the selected signals are green as the ambulance reaches them.

- **System Flow**

The overall flow of the proposed system is:
Route Selection
↓
Collect Signal Data
↓
Collect Cycle Times and Offsets
↓
Calculate Travel Times
↓
Form Modular Congruences
↓
Check Pairwise Coprime Condition
↓
Apply Chinese Remainder Theorem
↓
Calculate Starting Time
↓
Create Green Corridor
↓
Ambulance Reaches Hospital Faster

IMPLEMENTATION DETAILS

- **Introduction**

The implementation of the proposed system involves converting traffic signal information into mathematical equations and solving those equations using the Chinese Remainder Theorem.
The implementation does not require complicated hardware for demonstrating the mathematical concept. The major requirement is accurate information about the route, signal cycle times, signal offsets, and travel time.

Step 1: Collect Data

First, information about the ambulance route is collected.
The required data includes:

- Number of traffic signals.
- Cycle time of every signal.
- Starting offset of every signal.
- Travel time between signals.
- Required green time when the ambulance reaches each signal.

The PPT specifies route, signal cycle times, offsets, and travel times as the required input data.

Step 2: Form Congruences

After collecting the data, one modular congruence is created for every traffic signal.
For example:

$$x \equiv 2 \pmod 3$$
$$x \equiv 3 \pmod 4$$
$$x \equiv 1 \pmod 5$$

Each equation represents the required timing condition for one signal.

Step 3: Check Coprimality

Before applying the standard form of CRT, the cycle times should be pairwise coprime.
For example:

$$3,\ 4,\ 5$$

are pairwise coprime because:

$$gcd(3,4) = 1$$
$$gcd(4,5) = 1$$
$$gcd(3,5) = 1$$

Therefore, CRT can provide a unique solution modulo:

$$3 \times 4 \times 5 = 60$$

The PPT identifies pairwise coprimality as a condition for the standard CRT approach.

Step 4: Apply CRT
Consider the example:

$$x \equiv 2 \pmod 3$$
$$x \equiv 3 \pmod 4$$
$$x \equiv 1 \pmod 5$$

First calculate:

$$N = 3 \times 4 \times 5 = 60$$

Now:

$$N_1 = \frac{60}{3} = 20$$
$$N_2 = \frac{60}{4} = 15$$
$$N_3 = \frac{60}{5} = 12$$

The modular inverses are:

$$y_1 = 2,\ y_2 = 3,\ y_3 = 3$$

Therefore:

$$x = (2)(20)(2) + (3)(15)(3) + (1)(12)(3)$$
$$x = 80 + 135 + 36$$
$$x = 251$$

Now:

$$251 \mod 60 = 11$$

Therefore:

$$\boxed{x = 11}$$

If one unit represents 10 seconds:

$$11 \times 10 = 110 \text{ seconds}$$

Therefore, the calculated starting time is **110 seconds**.
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Step 5: Create Green Corridor
After calculating x, the ambulance is planned to start its journey at that time.
Because the modular equations were created according to the green-light requirements of the individual signals, the ambulance is expected to encounter the required green signals sequentially.
This coordinated sequence is called a **green corridor**.

- ## Algorithm

Algorithm: Ambulance Green Corridor Using CRT
Step 1: Select the ambulance route.
Step 2: Identify all traffic signals on the route.
Step 3: Record the cycle time of each signal.
Step 4: Record the travel time between signals.
Step 5: Calculate the required green-time remainder for every signal.
Step 6: Form one modular congruence for every signal.
Step 7: Check whether the moduli are pairwise coprime.
Step 8: Calculate N.
Step 9: Calculate each N_i.
Step 10: Find the modular inverse y_i.
Step 11: Apply the CRT formula.

RESULT AND ANALYSIS

- **Result**

The proposed mathematical model successfully demonstrates how the Chinese Remainder Theorem can be used for coordinating multiple traffic signals.
For the example consisting of three traffic signals:

$$x \equiv 2 \pmod 3$$
$$x \equiv 3 \pmod 4$$
$$x \equiv 1 \pmod 5$$

the CRT calculation gives:

$$x = 11$$

When one unit represents 10 seconds:

$$11 \times 10 = 110 \text{ seconds}$$

Thus, the ambulance should start at **110 seconds** according to the example. The complete timing pattern repeats every 60 units, which corresponds to 10 minutes when one unit equals 10 seconds.

- **Result Analysis**

The result demonstrates that a set of independent repeating traffic-signal cycles can be represented mathematically using modular congruences.
The CRT solution provides a common time that satisfies all the specified congruences.
The important outcomes are:

1. A common starting time can be calculated mathematically.
2. Multiple traffic signals can be considered simultaneously.
3. The solution is exact when the required CRT conditions are satisfied.
4. The approach can reduce unnecessary ambulance stops.
5. The method provides a mathematical basis for creating a green corridor.

CONCLUSION

The project **"Ambulance Routing and Traffic Signal Priority Using the Chinese Remainder Theorem (CRT)"** demonstrates an application of Discrete Mathematics and Number Theory to a real-world emergency transportation problem.

Traffic signals operate using repeating cycles, and these cycles can be represented using modular arithmetic. By representing every traffic signal as a modular congruence and applying the Chinese Remainder Theorem, it is possible to calculate a common starting time for an ambulance.

The proposed approach can help create a **green corridor**, allowing the ambulance to pass through selected traffic signals with fewer interruptions. This can potentially reduce travel time, minimize unnecessary stops, and improve safety during emergency transportation.

The project also demonstrates that mathematical concepts such as modular arithmetic and CRT are not limited to theoretical problems. They can be applied to practical engineering and computer-science problems such as traffic management, emergency vehicle routing, scheduling, and synchronization.

However, practical implementation requires accurate signal timing and travel-time information. Real-world traffic conditions may also change dynamically. Therefore, future versions of the system can integrate GPS, IoT-enabled traffic signals, real-time traffic information, and generalized CRT.

Overall, the Chinese Remainder Theorem provides a simple and mathematically precise foundation for developing an emergency traffic-signal synchronization system. The reference project concludes that CRT can provide an exact way to synchronize traffic signals and create a green corridor for situations where every second matters.

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